HW1



- Due on 10/16, pm 11:59
- 1. Image Read/Write
  - Write a program to implement BMP format image reading and writing.
    - The program should correctly process BMP images with different image sizes.
    - The following information must be obtained from the BMP header and printed to the console:
      - Image width
      - Image height
      - Bits per pixel
      - Row size before padding
      - Padding bytes per row
      - Actual row size including padding
      - Total image buffer size

HW1

- 2. Image Rotation (with BMP read/write)
  - Perform a 270-degree clockwise rotation on the input image to generate the output image.
  - Determine the output image size automatically.



- 3. Channel Interchange (with BMP read/write)
  - Interchange the channels of the rotated image, R ⇨ G, G ⇨ B, B ⇨ R.
  - E.g., The original R value becomes the new G value.
- 4. Measure the execution time
  - Describe What part of the program is included in the timing measurement
- *Hint*
  - You can resize the image as a small one to debug your program.

HW1

- Bonus
  - 1. Resize the image as double size and one-half size.
    - Hint: Up/Down sampling the pixels from the original image.
    - Repeat 1~3 for the resized image.
  - 2. Resize the image as 4096×4096
    - Hint: out of memory? malloc
    - Repeat 1~3 for the resized image.
  - 3. Compare and discuss
    - Repeat 1~3 for each image size: 512×512, 1024×1024, 2048×2048, 4096×4096. Measure and compare the execution time.
    - Analyze, explain, and verify the differences in execution time for various image sizes.

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- Requirements
  - Two Programs
    - C or C++ source code with .exe file (You are NOT allowed to use any library, such as OpenCV)
    - source code by using OpenCV
  - Report
    - Describe the employed source code editor and how to execute your program (input/interface/output)
    - Introduce your work, method, and discussions
    - With all of the images or results
    - **Explain the reason and rule of <u>padding</u>**
    - *AI Usage Declaration (tool/purpose/modification or none)*
    - *AI-generated text is strictly prohibited in the report*
  - Upload to i-school Plus